

ENTHUSIAST ADVANCE COURSE

PHASE : MEB-I (A)

TARGET : PRE-MEDICAL 2024

Test Type : MINOR

Test Pattern : NEET (UG)

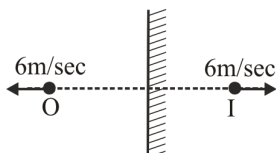
TEST DATE : 05-02-2023

ANSWER KEY

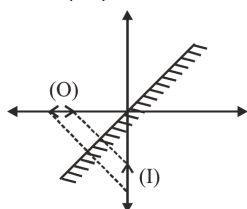
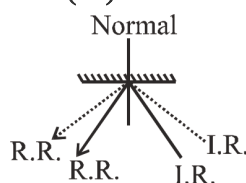
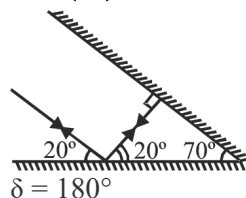
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A.	3	4	2	1	3	4	3	1	3	2	1	2	1	2	1	1	2	3	1	1	2	2	1	1	1	4	4	4	1	3
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	2	1	2	4	2	2	3	2	1	1	2	2	1	3	2	4	1	3	3	3	2	2	4	2	2	3	3	2	3	2
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
A.	3	3	2	4	2	1	3	4	2	3	1	2	1	1	1	4	3	2	4	1	3	4	2	2	4	1	4	4	3	2
Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
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Q.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
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A.	3	2	4	1	2	3	1	4	4	1	1	2	1	1	3	2	2	1	4	3	3	1	2	3	4	3	4	3	2	2
Q.	181	182	183	184	185	186	187	188	189	190	191	192	193	194	195	196	197	198	199	200										
A.	1	1	2	2	2	4	1	2	1	1	1	2	2	2	4	4	1	3	2	3										

HINT - SHEET**SUBJECT : PHYSICS****SECTION-A**1. **Ans (3)**

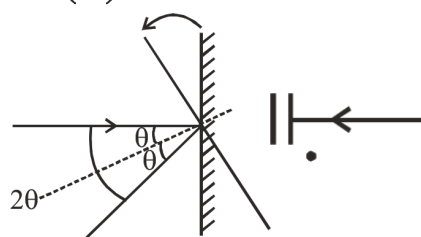
Relative speed of image w.r.t. object = $6 - (-6)$
 = 12 m/sec.

2. **Ans (4)**

$$\frac{5}{2} = 2.5 \text{ feet}$$

3. **Ans (2)**4. **Ans (1)**5. **Ans (3)**

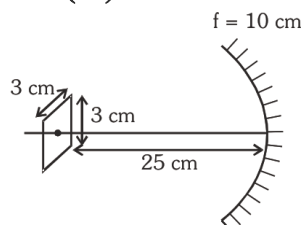
Ray retraces its path

6. **Ans (4)**

7. **Ans (3)**Here focal length = f and $u = -f$

On putting these values in

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v} \Rightarrow \frac{1}{f} = -\frac{1}{f} + \frac{1}{v} \Rightarrow v = \frac{f}{2}$$

8. **Ans (1)**

Side of square = 3 cm

$$A_0 = 3 \times 3 = 9 \text{ cm}^2$$

$$A_I = m^2 \times A_0$$

$$m = \frac{f}{f-u} = \frac{-10}{-10 - (-25)} = \frac{-10}{15} = \frac{-2}{3}$$

$$A_I = \left(\frac{-2}{3}\right)^2 \times 9 = 4 \text{ cm}^2$$

9. **Ans (3)**

$$\frac{AD}{40} = \frac{1}{4/3}$$

$$AD = 30 \text{ cm}$$

10. **Ans (2)**

Distance of object is equal to focal length

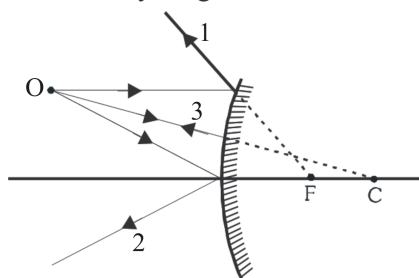
So, $u = \pm f$

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f} \Rightarrow \frac{1}{v} + \frac{1}{\pm f} = \frac{1}{f}$$

$$v = 2f, \infty$$

11. **Ans (1)**

Correct Ray diagram is :

12. **Ans (2)**

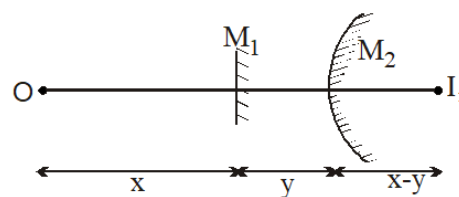
$$(v_I)_z = 2(v_M)_z - (v_O)_z$$

$$(v_I)_z = 2(0) - (3\hat{k})$$

$$(v_I)_z = -3\hat{k}$$

$$v_I = (v_I)_x + (v_I)_y + (v_I)_z$$

$$v_I = \hat{i} + 2\hat{j} - 3\hat{k}$$

13. **Ans (1)**

$$\frac{1}{x-y} + \frac{1}{-(x+y)} = \frac{1}{f} \Rightarrow f = \frac{x^2 - y^2}{2y}$$

Now, as this is convex mirror, f is (+)ve hence

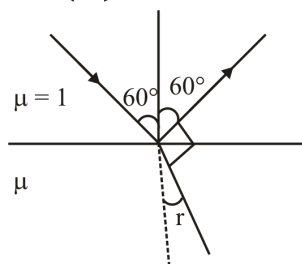
$$\text{magnitude of } f = \frac{x^2 - y^2}{2y}$$

14. **Ans (2)**

$$\text{Distance} = h_1 + \frac{h_2}{\mu}$$

15. **Ans (1)**

$$\lambda_{\text{medium}} = \frac{\lambda_{\text{air}}}{\mu} = \frac{6000}{1.5} = 4000 \text{ \AA}$$

16. **Ans (1)**

$$60 + r = 90^\circ$$

$$r = 30^\circ$$

$$1 \sin 60^\circ = \mu \sin 30^\circ$$

$$\mu = \sqrt{3}$$

17. **Ans (2)**From Snell's law $r = 2C'$

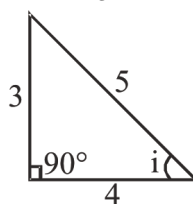
$$1.6 \times \sin i = 1 \times \sin r$$

$$1.6 \times \sin i = \sin 2C'$$

$$1.6 \times \sin i = 2 \sin i \times \cos i$$

$$\cos i = \frac{1.6}{2} = \frac{4}{5}$$

$$\sin i = \frac{3}{5}$$



$$i = \sin^{-1} \left(\frac{3}{5} \right)$$

19. **Ans (1)**

$$F = \frac{kQ^2}{r^2} \quad F_1 = \frac{kQ \cdot Q/2}{r^2} \Rightarrow F_1 = \frac{F}{2}$$

20. **Ans (1)**

$$F_{\text{net}} = 2\sqrt{2}F = 2\sqrt{2} \frac{kQq}{(a/\sqrt{2})^2}$$

$$= \frac{Qq\sqrt{2}}{\pi\epsilon_0 a^2}$$

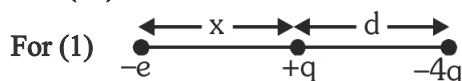
21. **Ans (2)**

$$\frac{KqQ}{R^2} = m\omega^2 R \Rightarrow T = 2\pi\sqrt{\frac{mR^3}{KqQ}}$$

22. **Ans (2)**

$$\text{For proton } e = \frac{2}{3}e + \frac{2e}{3} - \frac{e}{3} = uud$$

$$\text{For neutron } 0 = \frac{2}{3}e - \frac{e}{3} - \frac{e}{3} = udd$$

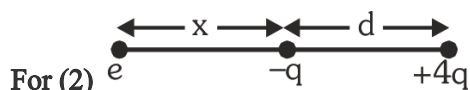
23. **Ans (1)**

Let the electron be held at a distance x from $+q$ charge.

$\therefore$ Distance of electron from $-4q$ charge $= (x + d)$

$$\text{For equilibrium } \frac{q(-e)}{4\pi\epsilon_0 x^2} = \frac{(-e)(-4q)}{4\pi\epsilon_0 (x+d)^2}$$

We can find value of x for which $F_{\text{net}} = 0$ which means that electron will be in equilibrium.



$$\text{For equilibrium } \frac{(-e)(-q)}{4\pi\epsilon_0 x^2} = \frac{(-e)4q}{4\pi\epsilon_0 (x+d)^2}$$

We can find value of x for which $F_{\text{net}} = 0$ which means that electron will be in equilibrium.

In case (3) and (4) the electron will not remain at rest, since it experiences a net non-zero force.

25. **Ans (1)**

$$q = ne = 10^{14} \times 1.6 \times 10^{-19} = 1.6 \times 10^{-5}$$

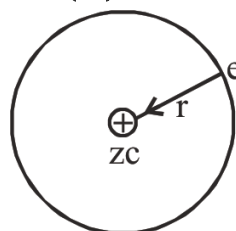
$$= 16 \times 10^{-6} = 16\mu\text{C}$$

26. **Ans (4)**

$$F \propto \frac{Q_1 Q_2}{r^2} \quad \dots\dots(i)$$

$$F' \propto \frac{2Q_1 \cdot 2Q_2}{(r/2)^2} \quad \dots\dots(ii)$$

$$\frac{F'}{F} = 16 \Rightarrow F' = 16F$$

27. **Ans (4)**

$$F = \frac{ke^2}{r^2} (-\hat{r}) = -\frac{Ke^2}{r^3} (\vec{r})$$

28. **Ans (4)**

Conservation of charge

29. **Ans (1)**

Refer module

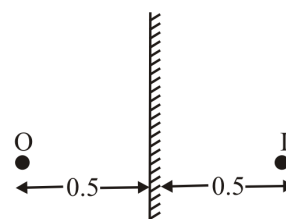
35. **Ans (2)**

$$P = \frac{a^3 b^2}{cd} \Rightarrow \frac{\Delta P}{P} \times 100$$

$$= \pm \left(3 \frac{\Delta a}{a} + 2 \frac{\Delta b}{b} + \frac{\Delta c}{c} + \frac{\Delta d}{d} \right) \times 100$$

$$= \pm (3 \times 1 + 2 \times 2 + 3 + 4) = \pm 14\%$$

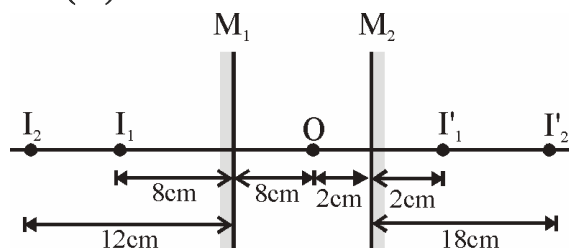
SECTION-B

36. **Ans (2)**

$$\text{distance between object and image} = 0.5 + 0.5$$

$$= 1 \text{ m}$$

37. Ans (3)



$$M_2 I'_1 = 2 \text{ cm}, M_2 I_1 = M_2 I'_2 = 18 \text{ cm}$$

$$M_2 I_2 = 10 + 12 = 22$$

38. Ans (2)

$$n_{ga} = \frac{3}{2}, n_{wa} = \frac{4}{3}$$

$$\therefore n_{gw} = \frac{n_{ga}}{n_{wa}} = \frac{3/2}{4/3} = \frac{9}{8}$$

39. Ans (1)

$$R = +20 \text{ cm}$$

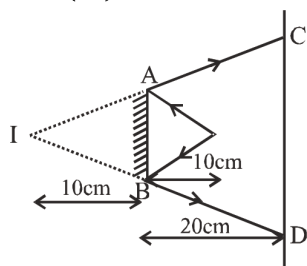
$$\Rightarrow f = \frac{R}{2} + 10 \text{ cm}; u = -14 \text{ cm}$$

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u} \Rightarrow \frac{1}{v} = \frac{1}{f} - \frac{1}{u}$$

$$= \frac{1}{10} + \frac{1}{14} = \frac{7+5}{70} = \frac{12}{70}$$

$$\therefore v = \frac{70}{12} \text{ cm}$$

40. Ans (1)


 $\Delta^s ABI \text{ \& } CDI \text{ are similar}$

$$\frac{AB}{CD} = \frac{10}{30} = \frac{1}{3}$$

$$CD = 3(AB) = 3(20)$$

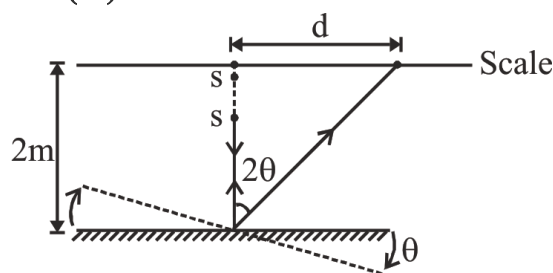
$$CD = 60 \text{ cm}$$

$$\text{time} = \frac{\text{Distance}}{\text{Speed}} = \frac{CD}{10} = \frac{60}{10} = 6 \text{ sec.}$$

41. Ans (2)

$$\text{Apparent height} = \mu h = \frac{4}{3} \times 24 = 32 \text{ cm}$$

42. Ans (2)



$$2\theta = \frac{d}{2}$$

$$2 \times 1.8^\circ = \frac{d}{2}$$

$$d = 4 \times 1.8^\circ$$

$$d = 4 \times 1.8 \times \frac{\pi}{180} = 0.1256 \text{ m}$$

43. Ans (1)

For concave mirror

$$u = +120 \text{ cm}$$

$$f = -60 \text{ cm}$$

$$v = \frac{uf}{u-f} = \frac{(120)(-60)}{120 - (-60)} = -40 \text{ cm}$$

$$m_1 = -\frac{v}{u} = -\frac{-40}{120} = +\frac{1}{3}$$

For convex mirror :

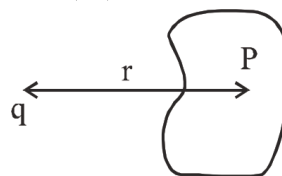
$$u = +(40 - 15) = +25 \text{ cm}$$

$$f = +30 \text{ cm}$$

$$m_2 = \frac{f}{f-u} = \frac{30}{30-25} = +6$$

$$m_{\text{net}} = m_1 \times m_2 = \frac{1}{3} \times 6 = +2$$

44. Ans (3)


 $\therefore$ Electric field inside conductor is zero

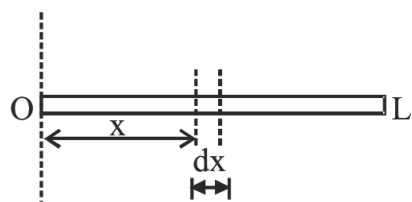
$$\therefore \vec{E}_q + \vec{E}_{\text{conductor}} = 0$$

$$E_{\text{conductor}} = \frac{-Kq}{r^2}$$

45. Ans (2)

Let us assume an element of length dx at a distance x from O.

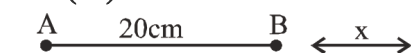
charge on element



$$dQ = \lambda dx = \frac{\lambda_0 x}{L} dx$$

$$\int_0^Q dQ = \int_0^L \frac{\lambda_0}{L} x dx \Rightarrow Q = \frac{\lambda_0}{L} \frac{L^2}{2}$$

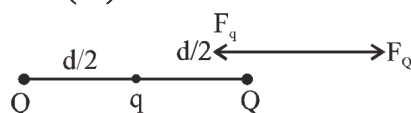
$$\Rightarrow Q = \frac{\lambda_0 L}{2}$$

46. Ans (4)

$$\frac{K 8 \times 10^{-6}}{(0.20 + x)^2} = \frac{K 2 \times 10^{-6}}{x^2}$$

$$\frac{2}{0.2 + x} = \frac{1}{x}$$

$$\Rightarrow x = 0.2 \text{ m} = 20 \text{ cm}$$

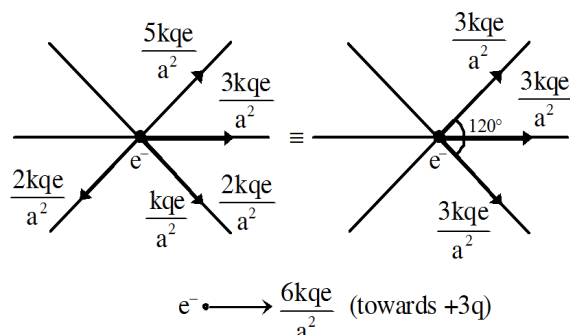
47. Ans (1)

For system to remain in equilibrium

$$F_Q = F_q \Rightarrow \frac{KQ^2}{d^2} = \frac{KqQ}{\left(\frac{d}{2}\right)^2}$$

$$\Rightarrow q = \frac{(-) Q}{4}$$

$\therefore F_q$ is attractive

48. Ans (3)**49. Ans (3)**

$$\text{Least count (LC)} = \left(\frac{b-a}{b} \right) \text{MSD ; here } a = 9,$$

$$b = 10 ; 1 \text{MSD} = 1 \text{mm}$$

$$\text{LC} = \left(\frac{10-9}{10} \right) 1 \text{mm} = 0.1 \text{mm}$$

There is zero in measurement.

As the zero of the vernier lies to the left of the main scale, it has negative zero error. For negative zero error, the coinciding division is to be read from right and it has 5th division coinciding from right.

$$\text{Thus zero error} = \text{ZE} = -5 \times 0.1 \text{ mm} = -0.5 \text{ mm} = -0.05 \text{ cm}$$

$$\text{Reading at measurement : MSR} = 8.5 \text{ cm}$$

$$\text{VSR} = 8^{\text{th}} \text{VSD} \times 0.1 \text{ mm} = 0.08 \text{ cm}$$

$$\text{Corrected reading} = \text{MSR} + \text{VSR} \times \text{LC} - \text{zero error}$$

$$= 8.5 \text{ cm} + 0.08 \text{ cm} - (-0.5 \text{ mm}) = 8.63 \text{ cm}$$

SUBJECT : CHEMISTRY**SECTION-A****51. Ans (2)**

The order of shielding effect of various orbital electrons is $s > p > d > f$. Due to the poor shielding effect of 4f- electrons in 5d-series elements, there is enhanced increase in effective nuclear charge. As a result of this the valence electrons are tightly bound with the nucleus and thus their removal require higher energy.

59. Ans (3)

electronegativity (approx. values)

$$\text{F} \quad 4.0$$

$$\text{Cl} \quad 3.16$$

$$\text{C} \quad 2.5$$

$$\text{P} \quad 2.19$$

$$\text{H} \quad 2.2$$

$$\text{H} - \text{P} = 2.2 - 2.19 = 0.01$$

60. **Ans (2)**

NCERT Pg. # 85

61. **Ans (3)**

NCERT Pg # 90

62. **Ans (3)**

NCERT Pg # 76

63. **Ans (2)**

NCERT Pg # 82

71. **Ans (1)**

From the given data half life is directly proportional to initial concentration or pressure therefore order of reaction is zero.

76. **Ans (4)**

$$\text{ROR} = \frac{-d[A]}{\frac{dt}{2}} = \frac{-[0.3 - 0.8]}{5 \times 2}$$

$$\frac{0.5}{5 \times 2} = \frac{0.1}{2} = 5 \times 10^{-2} \text{ M.min}^{-1}$$

NCERT Pg. # 98

77. **Ans (3)**

NCERT Pg. # 98

79. **Ans (4)**

NCERT Pg. # 103, 104, 112

81. **Ans (3)**

$$\log\left(\frac{K_2}{K_1}\right) = \frac{E_a}{2.303R} \left[\frac{1}{T_1} - \frac{1}{T_2} \right]$$

$$\log\left(\frac{4K_1}{K_1}\right) = \frac{E_a}{2.303 \times R} \left[\frac{1}{290} - \frac{1}{310} \right]$$

$$\frac{0.60 \times 2.303 \times R \times 310 \times 290}{20} = E_a$$

$$\frac{1.38 \times R \times 310 \times 290}{20} = E_a$$

82. **Ans (4)**

$$t_{99.9\%} = 10 \times t_{1/2}$$

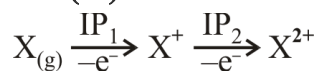
$$t_{1/2} = \frac{60}{10} = 6 \text{ min}$$

85. **Ans (4)**

$$\text{ROR (r)} = K = \frac{(\text{ROA}) H_2}{3}$$

$$(\text{ROR})_{H_2} = 3 \times 5 \times 10^{-4}$$

$$= 15 \times 10^{-4} \text{ mol L}^{-1} \text{ S}^{-1}$$

SECTION-B86. **Ans (1)**90. **Ans (2)**

NCERT Pg # 89

93. **Ans (2)**

$$= \frac{t_{75\%}}{t_{50\%}} = \frac{\frac{2.303}{k} \frac{100}{25}}{\frac{2.303}{k} \frac{100}{50}}$$

$$t_{75\%} = 2 \times 16 = 32$$

94. **Ans (3)**

Threshold energy = activation energy + energy possessed by reactant

$$= 32 + 18 = 50 \text{ K cal/mol}$$
95. **Ans (3)**

Ref. NCERT

99. **Ans (2)**

$$\frac{r_2}{r_1} = (2)^{\frac{40-10}{10}}$$

$$\frac{r_2}{r_1} = 2^3 = 8$$

100. **Ans (2)**

NCERT Pg. # 107

SUBJECT : BIOLOGY-II**SECTION-A**155. **Ans (2)**

NCERT XII, Pg # 23

156. **Ans (3)**

NCERT XII Pg.# 24

159. **Ans (4)**

NCERT-XII, Pg#21

160. **Ans (1)**

NCERT(XII) Pg # 21 (E), 22 (H)

171. **Ans (3)**

NCERT [XII] pg. 29

172. **Ans (1)**

NCERT (XI) Pg. # 85

175. Ans (4)

NCERT XII pg.# 34

178. Ans (3)

NCERT XII page 34-35, C & D are true

SECTION-B**187. Ans (1)**

NCERT-XII, Pg#23, Para.-II

188. Ans (2)

NCERT XII, Pg # 23

194. Ans (2)

NCERT (XII) Pg # 34

195. Ans (4)

NCERT-XII, Pg. # 34

197. Ans (1)

NCERT XII Pg.# 35

200. Ans (3)Microspore, Generative cell, synergid,
antipodal cell, vegetative cell of pollen